



Investigation Report

BM2210: Biomedical Device Design
Department of Electronic and Telecommunication Engineering
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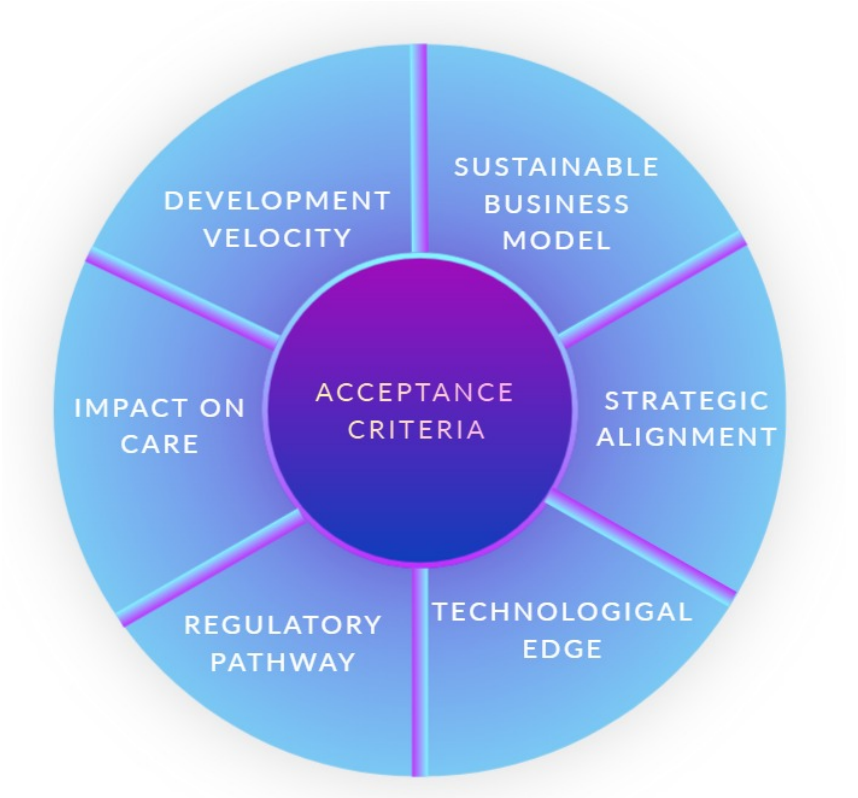
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1 Introduction

VitaSync is an emerging medical technology startup led by second-year undergraduates. Our mission is to bring bold, fresh ideas to the forefront of healthcare innovation and design the next generation of biomedical devices for a brighter future.

2 Our Strategic Focus



2.1 Our mission

VitaSync's mission is to transform the global healthcare landscape through innovative biomedical engineering. We design advanced, highly affordable devices that prioritize seamless usability, putting effortless, life-changing healthcare technology directly into the hands of the consumer.

2.2 Our Strengths and Capabilities

At VitaSync, we are transparent about our current standing as second-year undergraduates. We are still mastering the complexities of human anatomy and advanced electronic circuitry. However, our core strengths are rooted in out-of-the-box thinking, a strong grasp of foundational electronics, and a fierce dedication to continuous learning. We embrace these challenges as fuel for our growth. Through passion and perseverance, we are committed to pushing the boundaries of biomedical design to create highly impactful, affordable, and easy-to-use healthcare devices for our consumers.

2.3 Effect on affordability and accessibility of care

VitaSync is driven by a commitment to make quality healthcare accessible and affordable. We design low-cost, user-friendly biomedical devices that put the power of proactive health management directly into the hands of the consumer. By reducing complex barriers and high costs, our mission is to ensure that life-changing healthcare technology is within reach for everyone, regardless of their economic background.

2.4 Customers that need to be cultivated

The customers we aim to cultivate encompass a truly global market driven by a shared need for accessible, simple, and pioneering healthcare solutions. Because our primary focus is on extreme affordability, our core target demographic consists of everyday people worldwide who are seeking reliable, high quality medical devices to monitor and enhance their personal well being. Although we prioritize empowering the individual user directly, we are equally open to and capable of partnering with diverse organizations. This includes international healthcare providers, government health agencies, and global NGOs looking for scalable and cost effective hardware to improve patient care and public health outcomes in varied economic environments.

2.5 Market

Our market strategy at VitaSync is centered around addressing the universal need for more affordable and accessible electronics in the global healthcare sector. Although the global market for medical devices is expanding rapidly and is projected to exceed USD 1 trillion by 2034, price and accessibility hurdles continue to prevent essential medical technologies from reaching the vast majority of people around the world. By focusing our efforts on extreme affordability, our primary goal is to directly empower individuals across the globe who are seeking cost effective, high quality biomedical devices to manage and enhance their well being. Furthermore, while our core focus remains on the everyday consumer, our scalable market strategy seamlessly accommodates partnerships with a wide array of organizations, including international healthcare providers, NGOs, and government health agencies. This strategic, globally-minded approach allows us to leverage our engineering expertise to bridge the healthcare divide, significantly improving healthcare accessibility and positively impacting global health outcomes for all demographics.

2.6 Time and resources required to achieve the results

As engineering undergraduates, our research and development timeline is tightly constrained to a single academic semester (four months) to finalize our initial functional prototypes. Balancing this rapid development cycle alongside our rigorous coursework presents a unique challenge, which is further compounded by the limited availability of specialized biomedical components in the local supply chain and the financial pressures of fluctuating exchange rates. To navigate these procurement hurdles on a student budget, our approach relies on extreme engineering resourcefulness. By leveraging widely available standardized components, utilizing our university prototyping facilities, and translating our theoretical coursework into cost effective practical designs, our team is fully equipped to overcome these limitations and deliver on our mission of globally accessible medical devices.

3 Need Finding

3.1 Opportunity 1: Need for a medical device that can detect early warning signs of heat stroke

Resource Person: Dr. J.D.Fernando
Medical Officer at National Teaching Hospital, Kandy

The doctor referred to current heat stroke assessment practice, where sweat status and patient-reported symptoms are the main indicators clinicians work from.
Problems mentioned;

- **Single-Indicator Risk:** Flagged that sweat/anhidrosis alone isn't sufficient as a standalone diagnostic trigger. Highlighted considering the importance of parameters such as ECG and heart rate.
- **Symptom-based, Not Data-Based:** Current assessments are made with patient symptoms such as cardiac pain and sweating rather than continuous monitoring.
- **Field Monitoring Gap:** Identified human error, lack of continuous monitoring, and high cost of equipment as the current shortfalls in how field staff/site supervisors track workers for heat stress.

Proposed solution: a wearable device continuously tracking anhidrosis onset alongside temperature to give an early heat stroke warning before collapse.

- Explicitly recommended pairing sweat and temperature data with heart rate and respiratory rate to make the diagnosis more foolproof.

3.1.1 Need Statement

Problem: Heat stroke is a severe medical emergency characterized by thermoregulatory failure and dangerously high core temperatures. It is a significant risk for athletes, outdoor laborers, military personnel in extreme climates, and the elderly. A vital physiological warning sign is anhidrosis: the cessation of sweating due to severe dehydration or central nervous system failure. Rapidly drying skin amidst rising internal temperatures indicates imminent collapse, organ damage, or death without urgent intervention.

Population affected: Heat stroke disproportionately impacts individuals exposed to prolonged high temperatures and those with compromised thermoregulation. The most vulnerable populations include outdoor laborers in sectors like agriculture and construction, military personnel operating in tropical climates, and athletes engaging in intense physical exertion. Additionally, passive heat stroke poses a severe threat to elderly populations during sudden heatwaves, making accessible monitoring solutions critical for these high-risk groups.

Outcome: The planned end device is a compact wearable patch that integrates Ag/AgCl electrodes to measure Galvanic Skin Response (sweat conductivity) and an NTC thermistor to track skin temperature. Powered by a low-energy microcontroller, the device runs local sensor-fusion algorithms to detect the onset of anhidrosis.

"A way to continuously monitor the onset of anhidrosis (sweat cessation) in individuals exposed to high environmental heat or prolonged physical exertion in order to provide an actionable early warning of heat stroke before systemic physiological collapse occurs."

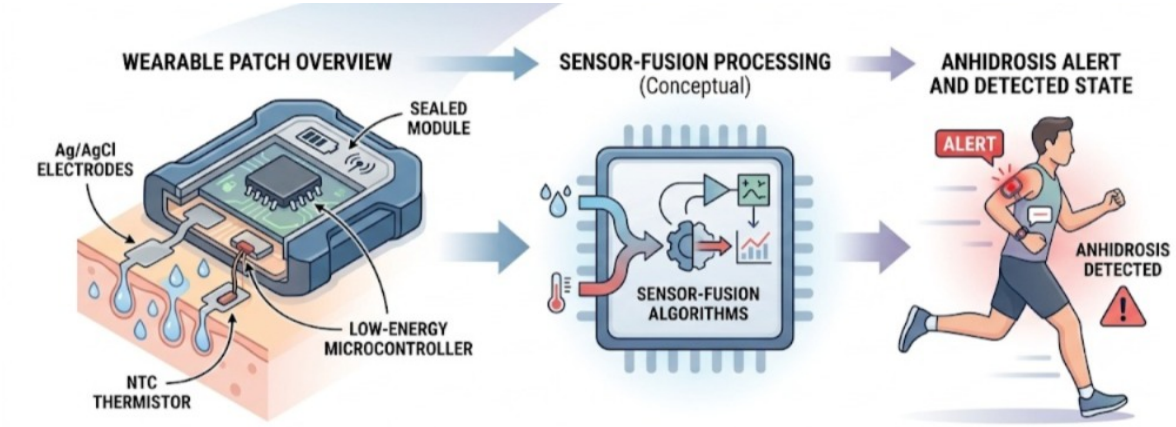


Figure 1: System Architecture

3.1.2 Disease State Fundamentals

What is the Disease?

- Heat stroke is a severe medical emergency characterized by abnormal thermoregulatory failure and dangerously high core temperatures. This condition often results from prolonged exposure to environmental heat or intense physical exertion.

Cause of the Disease:

- It typically occurs when the body's standard mechanisms for heat dissipation, such as sweating and cutaneous vasodilation, fail to keep up with metabolic or environmental heat loads.

Normal Anatomy Related to It and Abnormalities Caused by the Disease:

- A healthy individual maintains a stable internal core temperature through the hypothalamus. Heat stroke causes this central thermostat to malfunction, leading to cellular damage and systemic inflammation.

Symptoms and Signs:

- Early signs of heat stroke include severe headache, dizziness, nausea, and intense fatigue. As the condition worsens, critical indicators manifest as a core body temperature exceeding 40°C, altered mental states such as confusion or delirium, tachycardia, rapid breathing, and anhidrosis (the sudden cessation of sweating despite extreme heat).

Amount of People Facing This Issue:

- Heat stroke is a major global public health concern, particularly affecting outdoor laborers, military personnel operating in tropical climates, and athletes undergoing intense physical exertion. Additionally, vulnerable populations such as the elderly and individuals with compromised thermoregulation face high risk during sudden, severe heatwaves, which are increasing in frequency worldwide.

Cost of the Problem to the Health Care Sector:

- Management of heat stroke incurs significant economic and resource burdens on the healthcare sector due to the necessity of immediate emergency interventions, intensive care monitoring, and long-term treatment for potential multi-organ damage. Proactive, non-invasive ambulatory monitoring systems offer a substantial cost-saving opportunity by preventing critical physiological escalation and reducing emergency hospitalization rates.

3.1.3 Existing Solutions

Environmental Monitors (WBGT Meters):

- Devices that measure Wet Bulb Globe Temperature to gauge the heat stress of the environment.
- Drawbacks: They measure the surrounding environment, not the biological response of the individual worker, completely missing personal physiological failure.

Commercial Smartwatches (e.g., Garmin, Apple Watch):

- Track heart rate and estimate skin temperature.
- Drawbacks: They do not measure Galvanic Skin Response (GSR) or sweat conductivity in real-time, missing the critical anhidrosis indicator. They are also easily damaged on construction or military sites.

Ingestible Core Temperature Pills:

- Workers swallow a telemetry capsule that transmits core temperature to an external monitor.
- Drawbacks: Highly invasive, prohibitively expensive for daily industrial use, and entirely impractical for the general public or elderly populations.

Buddy System / Visual Observation:

- Relying on co-workers or caretakers to look for signs of confusion, flushing, or stumbling.
- Drawbacks: Highly subjective, prone to human error, and usually identifies the problem only after the individual is already in medical distress.

3.1.4 Stakeholder Analysis

Primary Users:

- Construction workers
- Agricultural laborers
- Military personnel
- Athletes, cricketers (Specially for day matches in high-heat areas). As overall outdoor sportsmen
- Elderly individuals in high heat areas

Decision Makers / Buyers:

- Occupational Health and Safety (OHS) managers
- Construction/mining firm executives
- Military procurement officers
- Elderly care facility administrators

Secondary Stakeholders:

- Insurance companies (seeking to reduce worker's compensation claims for heat-related injuries)
- Labor unions

3.1.5 Market Analysis

Target Market:

- The initial beachhead market for the start-up would be industrial construction firms and agricultural companies operating in equatorial and tropical zones, where labor safety regulations regarding heat exposure are becoming increasingly strict.

Market Expansion:

- Once proven in industry, the device can scale horizontally into the consumer elderly-care market and high-performance athletics.

Viability:

- The financial incentive for companies is massive. Heat-related hospitalizations result in extreme medical costs, halted production, and severe legal liabilities. By engineering a ruggedized, low-cost wearable, the start-up can offer a high-margin, highly scalable safety net that costs a fraction of a single worker's compensation claim.

3.2 Opportunity 2: Need for a medical device that can continuously track joint angles and Range of Motion (ROM) during physical therapy

Resource Person: Ms. Isuri Tharuka Liyanage
BSc (Hons) in Physiotherapy (Pdn)
Diploma in Exercise and Sports Sciences
Physiotherapist
National Hospital, Galle

The physiotherapist referred to the clinical assessment of joint Range of Motion (ROM), a critical metric for evaluating recovery in orthopedic and neurological rehabilitation patients. To track patient progress, clinicians currently rely on periodic in-clinic assessments; while the standard measurement tool is a manual goniometer, doctors often assess ROM simply by visually observing the patient's motions during routine visits.
Problems mentioned;

- **Infrequent Monitoring:** Assessment only happens during scheduled clinic visits, so any progress, stiffness, or lack of exercise adherence that occurs at home between appointments goes completely unrecorded until the next check-up.
- **Subjective Estimation:** Because clinicians frequently rely on visual observation to gauge joint angles rather than precise measurement tools, the assessment lacks strict objectivity and depends heavily on the examining clinician's experience and perspective.
- **Lack of At-Home Visibility:**While current in-clinic measurement methods are not physically uncomfortable for the patient, they are completely isolated to the clinical environment. This leaves a massive data gap, as therapists cannot measure or verify how the joint is moving when the patient is performing unsupervised exercises at home.

The proposed solution is to develop a wearable medical device that can continuously and automatically track joint kinematics and ROM during at-home physical therapy. Such a device could use an Inertial Measurement Unit (IMU) embedded in a low-cost sensor band to provide continuous, real-time monitoring of daily exercises between clinic visits, making compliance tracking more objective, seamless for the patient, and highly accessible outside of the traditional clinical setting.

3.2.1 Need Statement

Problem: Physical therapy and orthopedic rehabilitation rely heavily on the precise measurement of joint kinematics to evaluate patient recovery. Following procedures like total knee arthroplasties, ligament repairs, or during stroke rehabilitation, patients must achieve specific Range of Motion (ROM) milestones to prevent permanent joint stiffness and scar tissue formation. A massive gap in care occurs when patients return home; therapists currently have no objective way to measure if a patient is performing their prescribed exercises correctly, resulting in poor compliance, prolonged recovery times, and permanent mobility loss.

Population affected: The target population includes post-operative orthopedic patients (such as those recovering from total knee arthroplasty or ACL reconstruction), neurological rehabilitation patients (like stroke survivors), and individuals suffering from chronic joint diseases. Additionally, this heavily impacts physical therapists, physiotherapists, and orthopedic surgeons managing large caseloads of remote or outpatient individuals who require continuous monitoring.

Outcome: The planned end device is a low-cost, wearable sensor band that integrates an Inertial Measurement Unit (IMU), to track continuous joint kinematics and mechanical flexibility. Powered by a low-energy microcontroller, the device captures daily ROM data and transmits it via Bluetooth to a mobile application dashboard, providing therapists with objective clinical data on at-home exercise compliance.

"A way to continuously and automatically track joint angles and Range of Motion (ROM) during unsupervised at-home physical therapy exercises in order to provide physical therapists with objective, real-time compliance data and prevent long-term mobility loss in rehabilitation patients."

3.2.2 Disease State Fundamentals

What is the Disease?

- Impaired Range of Motion (ROM) and joint stiffness are severe complications following orthopedic surgeries, physical trauma, or neurological events, characterized by the inability to move a joint through its normal physiological arc.

Cause of the Disease:

- It typically occurs due to scar tissue formation (arthrofibrosis), severe muscle atrophy, prolonged immobilization after a fracture, or disrupted neuromuscular pathways, which restrict normal joint articulation and limit mechanical flexibility.

Normal Anatomy Related to It and Abnormalities Caused by the Disease:

- A healthy joint relies on smooth cartilage, flexible ligaments, and functioning muscles to achieve full extension and flexion. Prolonged immobility or incorrect rehabilitation causes the surrounding connective tissues to fibrose and tighten, drastically reducing the joint's mechanical flexibility and altering the patient's gait or movement mechanics.

Symptoms and Signs:

- Early signs include localized pain during movement, joint inflammation, and an objective, measurable decrease in the maximum flexion or extension angle of the affected limb when compared to the patient's baseline or their contralateral limb.

Amount of People Facing This Issue:

- Impaired mobility is a major global healthcare challenge, particularly affecting the aging population undergoing joint replacements, athletes recovering from sports injuries, and stroke survivors requiring intensive, long-term neuromuscular rehabilitation to regain independence.

Cost of the Problem to the Health Care Sector:

- Poor rehabilitation compliance incurs significant economic and resource burdens on the healthcare sector due to the necessity for extended physical therapy sessions, prolonged pain management medications, and costly revision surgeries required to physically break down scar tissue in permanently stiffened joints.

3.2.3 Existing Solutions

Manual Universal Goniometers:

- Plastic protractors used manually by therapists to measure joint angles during clinical visits.
- Drawbacks: Highly prone to inter-rater reliability issues (human error between different therapists), time-consuming to document manually, and completely useless for continuous at-home monitoring.

Smartphone-Based Pose Detection Apps:

- Applications utilizing mobile device cameras and machine learning algorithms to estimate joint angles during movement.
- Drawbacks: Requires the patient to properly set up a camera, ensure adequate lighting, and remain in the frame. It is often too technically burdensome for elderly patients and only captures data when actively recording, missing passive daily movements.

Digital Inclinometers:

- Advanced clinical tools that provide digital readouts of joint angles for highly precise measurements.
- Drawbacks: The high cost of importation and procurement makes them inaccessible for small, low-resource clinics. Furthermore, they are designed strictly for episodic clinical use rather than continuous, unsupervised at-home monitoring.

Buddy System / Visual Observation:

- Relying on co-workers or caretakers to look for signs of confusion, flushing, or stumbling.
- Drawbacks: Highly subjective, prone to human error, and usually identifies the problem only after the individual is already in medical distress.

3.2.4 Stakeholder Analysis

Primary Users:

- Post-operative orthopedic patients (knee, hip, shoulder)
- Neurological rehabilitation patients (stroke survivors)
- Physical therapists and physiotherapists

Decision Makers / Buyers:

- Clinic owners and hospital administrators
- Orthopedic surgeons prescribing post-op care regimens
- Tele-rehabilitation platform operators

Secondary Stakeholders:

- Medical insurance companies (seeking to verify exercise compliance before approving further physical therapy sessions)
- Caregivers and family members of elderly patients

3.2.5 Market Analysis

Target Market:

- The initial beachhead market for the device would be private physical therapy clinics and orthopedic hospital wards seeking to modernize their outpatient care and offer data-driven tele-rehabilitation services, particularly in regions where high-end biomechanical tracking equipment is cost-prohibitive.

Market Expansion:

- Once validated in clinical rehabilitation settings, the technology can scale horizontally into the sports medicine sector for high-performance athlete recovery, and eventually into the commercial fitness wearable market.

Viability:

- The financial incentive for clinics is the ability to handle larger outpatient caseloads with better, verifiable recovery outcomes. By engineering a device utilizing low-cost embedded electronics, the start-up can offer a highly affordable hardware solution paired with a high-margin Software-as-a-Service (SaaS) subscription model for clinic access to the patient data dashboard.

3.3 Opportunity 3: Need of a medical device that can identify early symptoms of CKDu

Resource Person: Dr Dinith Galabada
MBBS,MD (University of Colombo)
Consultant Nephrologist
Colombo South Teaching Hospital

Dr. Dinith Galabada Consultant Nephrologist at Colombo South Teaching Hospital, MBBS, MD (University of Colombo) Following consultation with a consultant nephrologist, the decision was made not to go forward with the bioimpedance-based early CKD screening project due to fundamental physiological and technical limitations.
Problems mentioned;

- **Unstable Fluid Ratios:** In Stages 1 and 2, Intracellular/Extracellular Water (ICW/ECW) ratios naturally fluctuate, making early baseline detection impossible.
- **Environmental Noise:** Heat stress, sweat, and routine exertion distort physical parameters, masking subtle renal shifts.
- **Sensor Limitations:**Available non-invasive sensors lack the required precision to isolate early disease, and custom sensor development is unfeasible.
- **Superior Clinical Standard:**While it is prone to give false negative results from time to time, early kidney damage is reliably identified via chemical/biological markers (protein in urine), not biophysical fluid distribution.

3.3.1 Need Statement

Problem: Chronic Kidney Disease (CKD) affects an estimated 10% of the Sri Lankan population, with recent population studies indicating a crude prevalence between 10.7% and 14.4% Chronic Kidney Disease of Unknown Etiology (CKDu) is a devastating, non-communicable public health crisis endemic to the agricultural dry zones of Sri Lanka (specifically the North Central, Eastern, and Uva provinces) and other tropical farming regions in South Asia

Population affected: Unlike conventional Chronic Kidney Disease (CKD), which is mainly caused by long term diabetes or hypertension, CKDu (Chronic Kidney Disease of Unknown Etiology) mainly effects male agricultural laborers of dry zone districts such as Anuradhapura,Polonnaruwa and Monaragala. While the exact cause of this condition is still under scientific debate , it is highly correlated to Agricultural pesticides contaminating groundwater with nephrotoxic agrochemicals or heavy metals . Furthermore , extreme dehydration and prolonged occupational heat stress is linked to CKDu.

Outcome: Enter text

"A way to non-invasively and rapidly screen for early-stage renal degradation and fluid shifts in low-income agricultural workers within tropical environments, in order to identify CKDu before irreversible Stage 3 damage occurs."

3.3.2 Disease State Fundamentals

What is the Disease?

- Chronic Kidney Disease (CKD) is a long-term, progressive condition characterized by structural or functional abnormalities of the kidneys that persist for more than 3 months. The condition is classified into five distinct stages based on the estimated Glomerular Filtration Rate (eGFR) and the presence of kidney damage markers (such as albuminuria):
  - Stage 1 & 2 (Early Stage): Normal or slightly reduced filtering function with underlying structural/tissue damage.
  - Stage 3 (Moderate Stage): Moderate reduction in filtering capacity

- Stage 4 & 5 (Severe / End-Stage Renal Disease - ESRD): Severe loss of function where the patient requires regular kidney replacement therapy (hemodialysis or organ transplantation) to sustain life.

A key geographic variant is Chronic Kidney Disease of Unknown Etiology (CKDu), which predominantly affects young, non-diabetic agricultural workers in tropical dry zones.

Cause of the Disease:

- Conventional CKD Risk Factors:
  - Diabetes Mellitus (Types 1 & 2)
  - Hypertension (High Blood Pressure)
  - Glomerulonephritis & Polycystic Kidney Disease
- CKDu Primary Etiologies (Agricultural Hotspots):
  - Occupational Heat Stress & Recurrent Dehydration: Working under severe ambient temperatures leads to chronic intracellular dehydration, sub-clinical acute kidney injury (AKI) episodes, and localized inflammatory stress.
  - Nephrotoxic Environmental Exposures: Long-term exposure to heavy metals (e.g., Cadmium, Arsenic) present in agrochemicals, hard groundwater, or poor-quality fertilizer runoff.

Normal Anatomy Related to It and Abnormalities Caused by the Disease:

- A healthy human possesses two retroperitoneal kidneys, each containing roughly 1 million structural units called nephrons.
  - The Glomerulus: A high-pressure capillary network that filters blood, allowing water, ions, and small metabolic waste molecules to pass through while keeping proteins and blood cells inside the bloodstream.
  - The Tubulointerstitial Space: A complex network of renal tubules surrounded by capillaries and interstitial fluid.
- CKD/CKDu DEGRADED NEPHRON STRUCTURE:
  - Blood Flow - Glomerulosclerosis - Tubulointerstitial Fibrosis - Protein Leakage & Fluid Retention
- Pathophysiological Abnormalities in CKD / CKDu
  - Glomerulosclerosis
  - Tubulointerstitial Fibrosis & Atrophy
  - Disruption of Fluid & Osmotic Homeostasis
  - Systemic Uremia

Symptoms and Signs:

- Early-Stage Signs & Biomarkers (Asymptomatic Phase):
  - Sub-clinical shifts in cellular fluid distribution (ICW/ECW imbalance)
  - Microscopic albuminuria or presence of renal injury markers (e.g., KIM-1, SDMA).
- Late-Stage Clinical Symptoms (Stages 4 & 5):
  - Peripheral & Facial Edema
  - Chronic Fatigue & Anemia
  - Uremic Symptoms
  - Oliguria / Anuria

Amount of People Facing This Issue:

- An estimated 788 million people have chronic kidney disease (CKD) worldwide, making up about 14.2% of all adults. Most of these individuals live in low- and middle-income countries.

Cost of the Problem to the Health Care Sector:

- Total direct annual healthcare expenditures for diagnosed CKD and Kidney Replacement Therapy (KRT) across major regions exceed $370 Billion USD. This is projected to surpass $400 Billion USD over the coming years.

3.3.3 Existing Solutions

Laboratory Serum Creatinine & Estimated Glomerular Filtration Rate (eGFR) Testing:

- An eGFR (estimated Glomerular Filtration Rate) test is a blood test that estimates how well your kidneys filter waste. It uses the level of creatinine or cystatin C in your blood to calculate a score.
- Drawbacks: Creatinine is a lagging biomarker; it typically does not rise significantly until more than 50% of total kidney function is already permanently lost. Routine checkups are needed and rural farmers therefore must commute to hospitals often to get checkups , which is inefficient.

Urine Dipstick Colorimetric Testing (Albuminuria Screening):

- A chemical reagent strip is dipped into a urine sample to check for the presence of proteins (albumin) visually , shown by the color range indicated by the results of the test
- Drawbacks: Standard dipsticks are highly insensitive to the specific pathology of CKDu. patients are regularly shown to have no protein level issues even when they are at level 1 or level 2 of Renal failure , leading to high amounts of False negative testings.

Clinical Renal B-Mode Ultrasound Imaging:

- A clinician uses an ultrasound machine to visually inspect the physical dimensions and echogenicity of the kidneys.
- Drawbacks: While highly accurate at detecting structural thinning of the renal cortex, ultrasound machines are expensive, power-hungry, and require highly specialized clinical personnel to operate. They cannot be deployed as a mass, door-to-door screening tool in rural, off-grid villages and because of these complications , many do not opt for screening while the disease is at early stages and not show any heavy symptoms

Mobile Health (mHealth) Risk-Assessment Questionnaires:

- Health workers use statistical models to predict a risk factor for individuals , depending on the different aspects of their health,age,background,occupation etc.
- Drawbacks: This approach relies purely on epidemiological correlation and pattern recognition rather than data from the patient's body, therefore missing early biological variations.

3.3.4 Stakeholder Analysis

Primary Users:

- Low-income rural rice farmers, manual agricultural laborers, and regional community health nurses who conduct field-level health camps.

Decision Makers / Buyers:

- The Ministry of Health (Sri Lanka), regional provincial health directors, non-governmental organizations (NGOs) focused on rural clean water/health initiatives, and international healthcare development agencies.

Secondary Stakeholders:

- Clinical engineering departments (responsible for maintaining rural equipment), regional nephrologists who receive patient referrals, and agricultural cooperatives seeking to protect their workforce.

3.3.5 Market Analysis

Target Market:

- The initial target market comprises the state-funded mobile health screening clinics operating within the North Central and Uva Provinces of Sri Lanka, covering a high-risk population of over 1.5 million individuals.

Market Expansion:

- Once validated, the device can scale into other globally recognized CKDu hot spots, including the coastal agricultural belts of Andhra Pradesh and Uddanam in India, rural farming valleys in Nepal, and sugar cane plantations across Central America

Viability:

- The financial incentive for government adoption is massive. A single hemodialysis session in a regional hospital costs the state significant funds, and a patient requires multiple sessions per week. A reusable, low-cost electronic instrumentation system that screens thousands of patients at a tiny fractional cost per scan pays for itself within weeks by shifting patient care from high-cost terminal management to ultra-low-cost early lifestyle, hydration, and medical interventions.

3.4 Opportunity 4: Need for a medical device to continuously monitor plantar pressure and temperature for diabetic foot ulcer risk

Resource Person: Dr. P.L.B.Nishantha
Medical Officer at Nationa Teaching Hospital, Kandy

The doctor referred to the clinical management of diabetic foot ulcers, a common and serious complication of diabetic peripheral neuropathy. To catch these ulcers before they form, clinicians currently rely on periodic in-clinic screening performed once every few months during routine diabetes follow-up visits.

The problems mentioned are;

- **Infrequent Monitoring:** Screening only happens during scheduled clinic visits, so any pressure damage or inflammation that develops between appointments goes completely undetected until the next check-up.
- **Reliant on Patient Reported Symptoms:** Because neuropathy removes the patient's ability to feel pain or discomfort, patients often cannot report early warning signs themselves, and damage is usually only found once it is already visible as an open wound.
- **Limited Objectivity:** Manual inspection depends on the examining clinician's experience and the ulcer already being visually apparent, so it does not capture the underlying pressure or temperature changes that precede tissue breakdown.

3.4.1 Need Statement

Problem: Diabetic peripheral neuropathy causes patients to gradually lose protective sensation in their feet, so they can no longer feel the repetitive pressure, friction, or minor trauma that would normally cause a healthy person to shift weight or stop walking. This unnoticed mechanical and thermal stress accumulates at specific pressure points on the sole of the foot until it breaks down the tissue and forms a diabetic foot ulcer. Because the patient cannot feel the damage occurring, ulcers are typically only discovered once they are already open wounds, at which point treatment is slow, costly, and carries a high risk of infection and amputation. Clinicians and patients currently have no continuous, objective way to detect the early, pre-ulcer warning signs during normal daily activity outside the clinic.

Population affected: The target population includes patients with diabetes mellitus (type 1 or type 2) who have developed peripheral neuropathy and/or peripheral artery disease, particularly those with a prior history of foot ulceration or amputation, who

are at the highest risk of recurrence. This also affects podiatrists, diabetes foot clinics, wound care centres, and remote patient monitoring programmes that are responsible for managing large caseloads of at-risk patients between infrequent in-person visits.

Outcome: The proposed end device is a low-cost, washable smart insole embedded with an array of force-sensitive pressure sensors and temperature sensors positioned at the anatomical sites most prone to ulceration (heel, metatarsal heads, and great toe). A microcontroller continuously samples both pressure and temperature at each site, fuses the two signals, and flags sustained high-pressure loading or abnormal temperature asymmetry between the two feet well before a wound becomes visible. Data is transmitted to a mobile app and clinician dashboard so that both patient and provider can intervene early.

"A way to continuously and passively monitor plantar pressure distribution and localized foot temperature during normal daily activity, in order to detect the biomechanical and inflammatory precursors of diabetic foot ulcers before tissue breakdown occurs, as compared to episodic clinical screening, single-biomarker devices, or full-time-consuming manual inspection. "

3.4.2 Disease State Fundamentals

What is the Disease?

- A diabetic foot ulcer is a full-thickness break in the skin, most commonly on the plantar (sole) surface of the foot, that develops in a person with diabetes as a result of repeated unnoticed mechanical stress combined with impaired healing capacity. It is the end-stage visible symptom of a much longer, silent process of nerve and tissue damage

Cause of the Disease:

- The underlying cause is diabetic peripheral neuropathy - nerve damage caused by prolonged high blood glucose levels, which progressively destroys the sensory nerves in the feet. This removes the patient's normal pain and pressure feedback, so repetitive loading at bony prominences (heel, metatarsal heads, toes) during walking is never relieved. Many patients also have peripheral artery disease, which reduces blood flow and further impairs the tissue's ability to repair itself once damage begins. Foot deformities (e.g., calluses, flat foot, prior amputation) that concentrate pressure onto smaller areas are a major additional risk factor.

Normal Anatomy Related to It and Abnormalities Caused by the Disease:

- In a healthy foot, sensory nerve endings continuously report pressure and pain, prompting reflexive weight-shifting long before tissue damage occurs, while adequate circulation supplies oxygen and nutrients for constant micro-repair of the skin and underlying tissue. In diabetic neuropathy, this feedback loop is broken: nerve conduction slows and eventually fails at the sensory endings in the feet, and the combination of unnoticed repetitive pressure, connective tissue changes, and reduced blood flow leads to skin breakdown, callus formation, and eventually deep tissue necrosis at the affected site.

Symptoms and Signs:

- Because the disease is defined by loss of sensation, the earliest signs are typically not felt as pain by the patient. The two most clinically important pre-ulcer warning signs - established in podiatric literature - are localized plantar pressure sustained above safe thresholds during weight-bearing activity, and a temperature difference of roughly 2.2°C (4°F) or more between corresponding sites on the two feet, which indicates subclinical inflammation forming beneath the skin days to weeks before a wound becomes visible.

Amount of People Facing This Issue:

- A 2017 systematic review and meta-analysis estimated global diabetic foot ulcer prevalence at 6.3% of people with diabetes, with the lifetime risk of a person with diabetes developing a foot ulcer estimated at between roughly 19% and 34% across different reviews. Around 18.6 million people worldwide are affected by a diabetic foot ulcer in a given year, including approximately 1.6 million in the United States. Diabetic foot ulcers precede an estimated 80–85% of non-traumatic lower-limb amputations in people with diabetes, and the International Diabetes Federation has stated that a limb is lost globally to diabetes roughly every 30 seconds. Recurrence is also extremely common: up to 40% of healed ulcers recur within one year, and up to 65% recur within five years.

Cost of the Problem to the Health Care Sector:

- Diabetic foot ulcers and their complications are among the costliest complications of diabetes. Direct treatment costs in the United States alone are estimated at $9–13 billion annually, and when broader foot-related complications are included, roughly one in every three dollars spent on diabetes care in the US - over $80 billion annually - goes toward managing foot complications. A 2020 study with Kaiser Permanente found that a remote temperature-monitoring programme reduced costs by approximately $12,000 per patient per year, illustrating both the scale of the cost burden and the potential savings from earlier detection. Outcomes are also severe: the five-year mortality rate following a diabetic foot ulcer is approximately 30%, rising to over 70% for patients who undergo a major amputation.

3.4.3 Existing Solutions

Manual Screening (Monofilament Testing and Visual Inspection):

- The current clinical standard is periodic in-clinic screening using a Semmes-Weinstein monofilament to test for loss of protective sensation, combined with visual foot inspection by a podiatrist or diabetes educator.

- Drawbacks: This only captures a single snapshot in time, typically once every few months. It relies entirely on the patient or clinician noticing a problem during a brief appointment and provides no information about what happens to the foot during the weeks between visits, when the damage actually accumulates.

Orpyx SI Sensory Insoles:

- A custom-fabricated smart insole system that continuously measures plantar pressure, temperature, step count, and wear time, and gives the patient real-time audio/visual alerts when pressure at a given site exceeds a safe threshold for too long. A randomised proof-of-concept trial published in The Lancet Digital Health found an 86% reduction in ulcer recurrence at monitored sites among patients who used the system consistently.
- Drawbacks: The insoles are custom-fabricated for each patient's foot, which adds cost and lead time, and the product is positioned as a subscription-based remote patient monitoring service bundled with a dedicated care team rather than a simple standalone device, making it inaccessible for lower-resource clinics or price-sensitive patients.

Podimetrics SmartMat:

- An FDA-cleared floor mat that the patient stands on for 20 seconds once a day; it measures a full thermal map of both feet and flags temperature asymmetry between the left and right foot. In a 34-week clinical study, it detected 97% of developing ulcers an average of five weeks before they were noticed clinically.
- Drawbacks: It only measures temperature, not pressure, so it captures inflammation but not the mechanical loading that causes it. It is also a once-a-day snapshot rather than continuous monitoring, and it is entirely dependent on the patient remembering to stand on the mat every single day.

Siren Socks and Sensoria Socks:

- Siren Socks weave temperature sensors directly into the sock fabric for continuous, passive temperature monitoring without requiring a daily deliberate action. Sensoria Socks instead use textile pressure sensors woven into the sole, marketed mainly for gait and fitness analytics.
- Drawbacks: Each of these products captures only one of the two clinically relevant biomarkers (Siren: temperature only; Sensoria: pressure only, and marketed for fitness rather than clinical ulcer prevention), so neither gives a complete picture on its own.

3.4.4 Stakeholder Analysis

Primary Users:

- Patients with diabetes and diagnosed peripheral neuropathy, particularly those with a prior history of foot ulceration or amputation
- Podiatrists and diabetes foot-care clinicians who manage these patients between visits

Decision Makers / Buyers:

- Diabetic foot and wound care clinics, and hospital podiatry departments
- Integrated health systems and remote patient monitoring (RPM) programmes that already reimburse temperature-monitoring devices

- Health insurers and payers operating value-based care contracts, who bear the downstream cost of amputations

Secondary Stakeholders:

- Caregivers and family members of patients with reduced mobility or cognitive load
- Diabetes educators and home health nurses who support patient self-management
- Orthotics and prosthetics providers who fit custom footwear for at-risk patients

3.4.5 Market Analysis

Target Market:

- The initial beachhead market is outpatient diabetic foot and wound care clinics, and remote patient monitoring programmes within larger integrated health systems, where a reimbursement pathway for remote temperature/pressure monitoring already exists. These settings already manage the highest-risk patients where the clinical and financial case for prevention is strongest.

Market Expansion:

- Once validated for secondary prevention (patients with a prior ulcer), the device can expand to primary prevention for the much larger population of diabetic patients with neuropathy but no prior ulcer. Longer-term, the same dual pressure-and-temperature sensing platform could extend to other neuropathic conditions or general elderly fall-risk and gait monitoring, broadening the addressable market beyond diabetes alone.

Viability:

- The clinical and financial case is strong: avoided amputations and hospitalisations generate large, well-documented cost savings (on the order of $12,000 per patient per year in one published study), which supports a hardware-plus-subscription business model similar to Podimetrics' and Orpyx's. By combining both biomarkers that competitors currently offer separately, and by targeting a lower manufacturing cost than a fully custom-fabricated insole, a start-up could differentiate on affordability and completeness of data rather than competing head-on with an established, clinically-proven incumbent like Orpyx.

3.5 Opportunity 5: Non-Invasive Dental Caries Detection via Optical Fluorescence Imaging

Resource Person: Dr.H.G. Subhashitha Wijerathne
Base hospital Galgmuwa

The doctor referred to current caries screening: visual-tactile inspection with an explorer probe and conventional intraoral X-rays.

- **Late Detection:** X-rays only show decay after 30–40% of the mineral matrix is already lost.

- **Radiation Exposure:** : Frequent X-ray screening isn't suitable for children or pregnant patients.
- **Probe Risk and Subjectivity:** Explorer probes can puncture remineralizable enamel and spread bacteria, and readings vary by dentist experience.

Proposed solution: a handheld 405nm fluorescence wand that lights up healthy enamel green and bacterial activity red, giving a numerical Caries Score without radiation or probing.
Concerns:

- **Fluorescence isn't decay-specific:** Stains, plaque, calculus, and certain restorative materials can also fluoresce, risking false positives if the algorithm isn't well-tuned.
- **Surface-only limitation:** Optical fluorescence reads the enamel surface; it may still miss deep or interproximal (between-teeth) decay that X-rays can catch structurally.
- **No replacement for X-ray in some cases:** Bone loss, abscesses, and impacted teeth still require radiographic imaging.
- **Workflow fit:** Adds an extra scanning step to a routine exam; adoption depends on it being fast enough not to disrupt patient throughput in high-volume clinics.

3.5.1 Need Statement

Problem: Standard clinical diagnostics rely heavily on visual tactile inspection with mechanical explorer probes and conventional intraoral X rays, both of which frequently fail to catch early stage subsurface enamel demineralization and active bacterial colonization beneath the surface or in tight interdental spaces. Mechanical probes risk physically damaging fragile, remineralizable enamel, while X rays expose patients to ionizing radiation and only reveal density loss after 30 to 40 percent of the mineral matrix is already destroyed. Without an objective, real time diagnostic tool, clinicians miss the critical window for non invasive preventive care, leading to irreversible cavitation, painful drilling, and expensive restorative procedures.

Population affected: The primary population includes pediatric patients, pregnant women, and individuals seeking radiation free dental care, as well as general dental patients vulnerable to early subsurface or interdental decay. It directly impacts general dentists, pediatric dentists, dental hygienists, and public health community clinics managing high patient volumes with limited access to high cost diagnostic technology.

Outcome: A low cost, handheld intra-oral wand integrating a 405 nanometer narrow band LED, custom optical long pass filtering, and a high resolution micro camera module paired with open source image processing. The device captures real time optical fluorescence, highlighting healthy enamel in green and bacterial porphyrins in red, and computes a numerical Caries Score from 1 to 10 to quantify early demineralization and track lesion progression over time without using ionizing radiation.

"A way to continuously and non invasively detect and quantify early stage enamel demineralization and bacterial caries activity in dental patients, without using ionizing radiation, in order to enable early preventive intervention

before irreversible structural cavitation occurs, compared to visual tactile inspection, conventional intraoral X rays, or high cost commercial fluorescence systems."

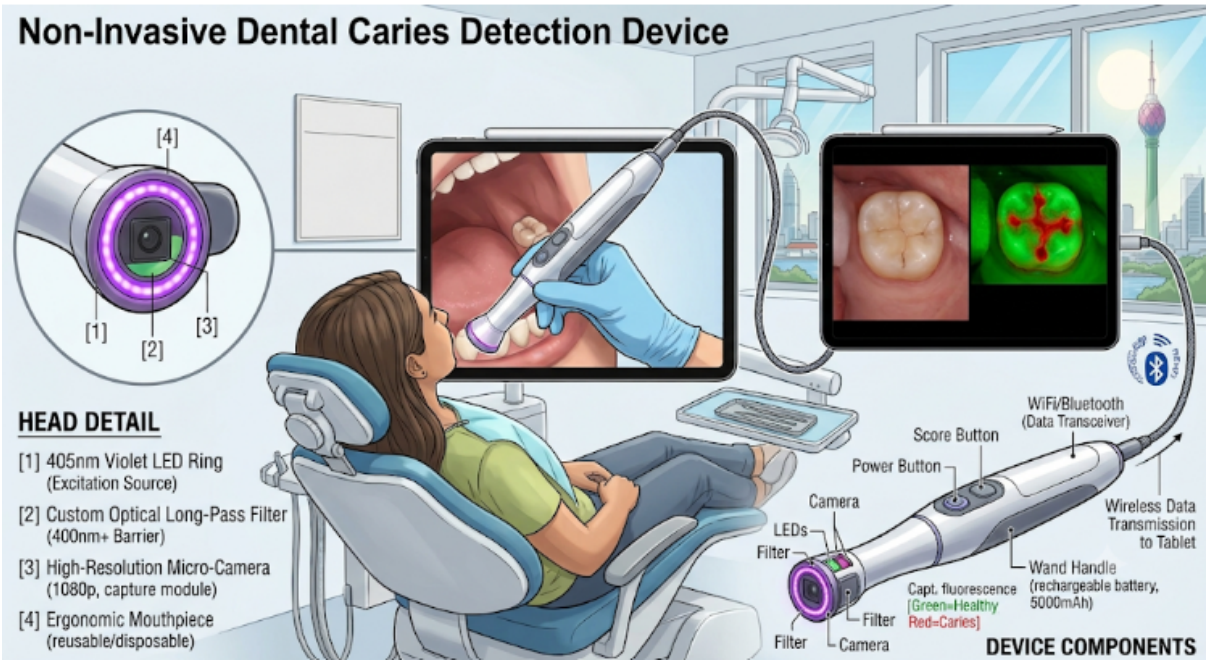


Figure 2: System Architecture

3.5.2 Disease State Fundamentals

What is the Disease?

- Dental caries (tooth decay) is a chronic oral disease characterized by the localized destruction of susceptible dental hard tissues (enamel and dentin) by acidic byproducts from bacterial fermentation of dietary carbohydrates.

Cause of the Disease:

- Acid-producing bacteria principally Streptococcus mutans and Lactobacillus species colonize the tooth surface within bio-films (plaque). Over time, sustained metabolic acid production drops local pH, dissolving the hydroxyapatite mineral matrix of enamel.

Normal Anatomy Related to It and Abnormalities Caused by the Disease:

- Healthy enamel is a highly mineralized, translucent structure. Demineralization increases enamel porosity, causing loss of translucency (early "white spot" lesions) and eventually leads to structural collapse (cavitation), dentin exposure, and pulpal inflammation.

Symptoms and Signs:

- Early signs include asymptomatic chalky white spots or micro-porosities on enamel. Advanced stages present with visible dark cavitated lesions, localized pain, temperature sensitivity, and severe infection.

Amount of People Facing This Issue:

- Dental caries is one of the most prevalent chronic conditions globally, affecting over 2 billion people worldwide across pediatric, adult, and geriatric populations.

Cost of the Problem to the Health Care Sector:

- Untreated caries places a massive financial burden on global healthcare systems through high restorative costs (fillings, crowns, root canals), lost productivity, emergency dental visits, and avoidable general anesthesia cases in pediatric patients.

3.5.3 Existing Solutions

Conventional Intraoral Dental Radiographs (X-Rays):

- Enter text

- Drawbacks:Exposes patients to ionizing radiation, making frequent screening unsuitable for children and pregnant mothers. Additionally, X-rays only reveal density changes after significant mineral loss (30 - 40%) has already occurred.

Visual-Tactile Inspection (Explorer Probe & Mirror):

- Enter text

- Drawbacks:Highly subjective and dependent on the dentist's experience. Sharp mechanical probes can puncture soft, remineralizing enamel lesions, inadvertently spreading bacterial infection into deeper layers.

High-End Commercial Optical Systems (e.g., QLF, KaVo DIAGNOdent):

- Enter text

- Drawbacks: Extremely expensive (3,000 - 10,000), making them unaffordable for standard private practices and community clinics. They rely on proprietary, closed-source software and bulky desktop setups that lack portability and seamless mobile integration.

3.5.4 Stakeholder Analysis

Primary Users:

- General dentists, pediatric dentists (pedodontists), dental hygienists, and clinical assistants.

Decision Makers / Buyers:

- Dental clinic owners, hospital dental department heads, and public health procurement officers.

Secondary Stakeholders:

- Dental academic researchers, pediatric health advocacy groups, and health insurance providers (interested in preventive care cost reduction).

3.5.5 Market Analysis

Target Market:

- The immediate beachhead market includes private dental clinics, community oral health centers, and pediatric dental practices seeking low-cost, radiation-free diagnostic tools.

Market Expansion:

- Starting with pediatric and pregnant patient care, the product expands into low-cost public health clinics and routine dental practices, eventually scaling into teledentistry platforms for remote patient monitoring.

Viability:

- By using off-the-shelf 405 nm narrow-band LEDs, custom optical long-pass filters, and high-resolution micro-camera modules (ESP32-CAM / Raspberry Pi setup) paired with open-source image processing (OpenCV), the total Bill of Materials (BOM) can be reduced to under USD 50. This drastic cost reduction makes digital fluorescence diagnosis universally accessible to everyday clinics.

4 Need Selection

Need	Estimated Market	Patient Impact	Providers imapct	Feasibility Index	Co-owners preferances	Total	Rank
	Out of 5	Out of 5	Out of 5	Out of 5	Out of 5	Out of 25	
A medical device that can continuously track joint angles and Range of Motion (ROM) during physical therapy	3	3	3	5	3	17	2
A medical device that can detect early warning signs of heat stroke	4	4	4	3	5	20	1
A medical device that can identify early symptoms of CKDu	2	4	3	2	3	14	4
A medical device to continuously monitor plantar pressure and temperature for diabetic foot ulcer risk	3	4	2	3	3	15	3
Non-Invasive Dental Caries Detection via Optical Fluorescence Imaging	2	2	3	3	2	12	5

5 Conclusion

In conclusion we have decided that the opportunity 01, **the need for a medical device that can detect early warning signs of heat stroke** to be the need that suits our objectives and the most viable product that we can work on.

Need Criteria: To effectively address the critical risk of heat stroke, the proposed biomedical device must fulfill several essential need criteria derived from the limitations of current solutions. Primarily, it must provide continuous, real-time, and objective monitoring of anhidrosis using physiological markers, ensuring an actionable early warning before systemic collapse occurs. The solution must be entirely non-invasive to guarantee high daily compliance and accessible enough for widespread use among vulnerable populations. Finally, the system must be highly cost-effective and scalable to appeal to occupational health and safety decision makers seeking an alternative to expensive medical interventions and inaccurate observational methods.

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